University of Studies of Milan

Computer Science Department Giovanni degli Antoni

Online Game Design

TECHNICAL DESIGN DOCUMENT

CATNIPPED!

ACADEMIC YEAR 2022-2023

Luca Ghisi

990732

Game Programmer, Infrastructure Engineer

luca.ghisi@studenti.unimi.it

Giovanni Morlacchi

27322A

Game Designer, Level Designer

giovanni.morlacchi@studenti.unimi.it

Pietro Arsi

15186A

Game Designer, Music and Art Director

pietro.arsi@studenti.unimi.it

Contents

Changelog

1 Project Goal **1**

2 Provided Services **1**

3 Client Side **1**
3.1 Hardware Requirements 2
3.2 Software Requirements 2

4 Workload Estimation **2**
4.1 Overcooked! 2 3
4.2 Moving Out 4
4.3 Our Estimation 5

5 General Architecture **6**
5.1 Front-end 6
5.1.1 Steam 6
5.1.2 Discord 7
5.1.3 Game Client 7
5.2 Back-end 7
5.2.1 Relay Service 7
5.2.2 Lobby 9
5.2.3 Authentication 10
5.2.4 Analytics 10
5.3 AWS 11
5.4 Workload Capacity 13
5.4.1 Front-end 13
5.4.2 Back-end 13

6 External Services **14**
6.1 Advertisement on social media 14
6.2 Payments 14
6.3 Achievements & Leaderboards 14
6.4 Authentication 15
6.5 Legal support for terms of service 15

7 Connection **15**
7.1 Global Infrastructure 15
7.2 Network Requirements 15

8 Gantt **17**

9 Delivery **18**
9.1 Estimated Delivery Time 18
9.2 Delivery Platform 18
9.3 Delivery Methodology 18
9.4 Post launch 18

10 Cost Estimation **19**
10.1 Team 19
10.2 Location 20
10.3 Hardware 20
10.4 Software 20
10.5 Unity Gaming Service 20
10.6 Advertisement 21
10.7 Total 21

Changelog

Users	Date	Description
Giovanni Morlacchi	27/04	Document Created
Luca Ghisi	28/04	Added Project Goal
Giovanni Morlacchi	30/04	Revised Project Goal
Luca Ghisi	01/05	Added Provided Services
Giovanni Morlacchi	02/05	Revised Provided Services
Luca Ghisi	02/05	Added Client side
Giovanni Morlacchi	04/05	Revised Client Side
Luca Ghisi	05/05	Added Workload Estimation
Giovanni Morlacchi	07/05	Revised Workload Estimation
Luca Ghisi	08/05	Added General Architecture
Giovanni Morlacchi	10/05	Revised General Architecture
Luca Ghisi	12/05	Added External Services Added Connection
Luca Ghisi	13/05	Improved General Services Improved Back-end
Giovanni Morlacchi Pietro Arsi	14/05	Revised Front-end
Luca Ghisi	14/05	Improved Connection Added Cost Estimation
Giovanni Morlacchi	14/05	Added Gantt
Giovanni Morlacchi Pietro Arsi	15/05	Revised Backend
Giovanni Morlacchi Pietro Arsi	15/05	Improved and Revised Relay Service Improved and Revised Lobby Improvedla and Revised Authentication
Pietro Arsi	15/05	Added External Services
Luca Ghisi	16/05	Improved General Infrastructure
Pietro Arsi	16/05	Improved Cost Estimation Improved AWS
Giovanni Morlacchi	16/05	Added Front Cover Revised AWS
Giovanni Morlacchi Luca Ghisi Pietro Arsi	17/05	General Revision Stylistic and Grammatical Improvements
Giovanni Morlacchi Pietro Arsi	18/05	Improved and Revised Cost Estimation
Giovanni Morlacchi Luca Ghisi	18/05	Final Revision

1 Project Goal

Catnipped! is a cooperative construction and management game that offers strategic gameplay. In *Catnipped!* a group of 4 players conquers a series of short levels-based challenges. The goal is to provide 24/7 access to the game for up to 4 players, in the first prototype this will be done for PC only.

2 Provided Services

For a complete player experience, we want to provide other services supporting the game. These services shall consist of:

- *Discord Server.* Dedicated to building a community, releasing updates, providing customer support, and a dedicated section to let players organize private or public lobbies
- *Steam.* The platform we will be using to distribute the game and its additional content
- *Social Media.* The most important platforms will be Facebook, Twitter, Instagram, TikTok and Youtube. These shall be used to help build a community around the game and share information about new upcoming updates and events
- *Updates and DLCs. Catnipped!* shall be supported after its first release, both in terms of bug-fixing updates, events, and DLCs, adding different modes, levels, and skins

3 Client Side

Catnipped! aims to be a light game able to run on most PCs. The requirements have been estimated based on similar games (Section 3.4 of the GDD), some of which are also present in Section 4. Although *Overcooked! 2* is in the top performers' Section of the GDD because of its player base, this Section has also been influenced by *Overcooked: All You Can Eat* since it came out more recently.

3.1 Hardware Requirements

	Minimum	Recommended
Processor	Intel Core 2 Quad Q6600 AMD Phenom II X3 720	Intel Core 2 Quad Q6600 AMD FX-4300
Memory	2 GB	4 GB
VRAM	1 GB	2 GB
Graphics	GeForce 8800 GT AMD HD 6850 Intel HD Graphics 4400	Nvidia GeForce GTX 660 Radeon HD 7510
Sound card	DirectX 11 Compatible Card	DirectX 11 Compatible Card
Storage	4GB	10 GB
Connection	Broadband Internet Connection	Broadband Internet Connection
Network	10 Mb/s	20 Mb/s

Table 1: Harware requirements for *Catnipped!*

3.2 Software Requirements

	Minimum	Recommended
OS	Windows 7 (64-bit)	Windows 10 (64-bit)
Client Software	Steam	Steam
DirectX Version	11	11

Table 2: Software Requirements for *Catnipped!*

4 Workload Estimation

Thanks to the data provided by SteamDB, a feasible workload can be estimated by analyzing the main competitors that have similar features to our game. Data from this Section was obtained using SteamDB. In this Section, we will present the two main competitors: *Overcooked! 2* and *Moving Out*. These games have been chosen because of their similarities to *Catnipped!*, both from the graphics and gameplay standpoints. Focusing on *Overcooked! 2*, although the game shares a lot of similarities with ours, other versions of it have been considered for different reasons, discussed in Sections 3 and 4.1.

4.1 Overcooked! 2

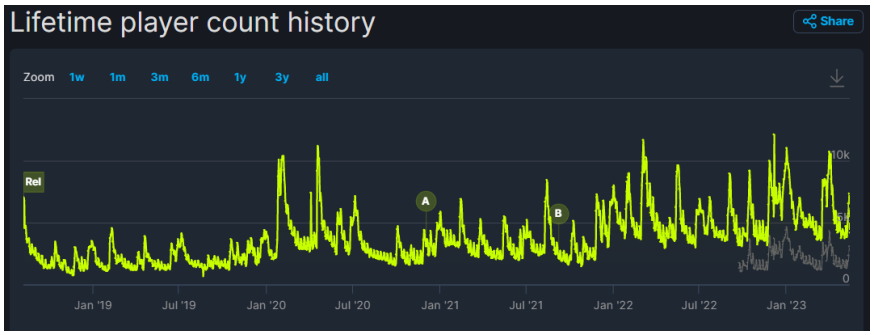


Figure 1: *Overcooked! 2* player count

Developed by Ghost Town Games Ltd and published by Team17 Digital Ltd, *Overcooked* is a co-op game, where one to four players have to prepare, cook and serve different orders before the buying customers leave. *Overcooked! 2* takes the formula of its predecessor and adds the possibility to play both as a couch game and as an online cooperative game. This is the primary reason why we have chosen the second chapter of the game as our comparison. It should be noted that *Overcooked: All You Can Eat*, a remaster of both *Overcooked* and *Overcooked! 2*, was released in 2020 for next-gen consoles.

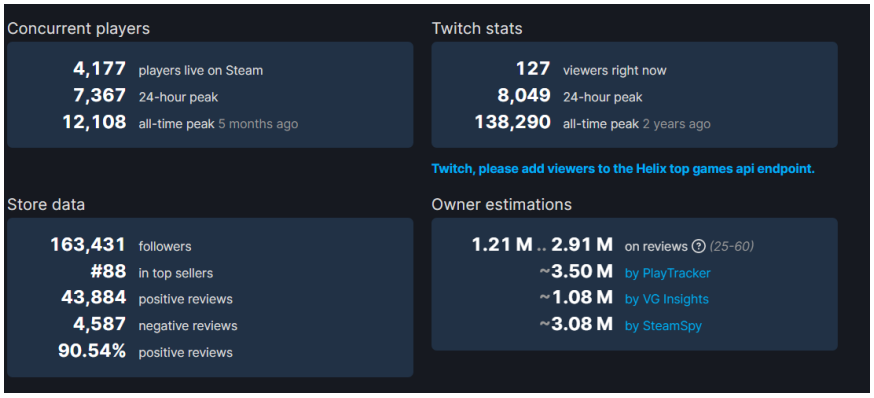


Figure 2: *Overcooked! 2* data for SteamDB

At launch the game had 6760 Concurrently Connected Users (CCU) and, as can be seen in Figure 2, it reached a peak of 12108 players. This is also thanks to the success of its predecessor, *Overcooked*, a local multiplayer coop couch game. This game (Figure 3), had 500 CCU at launch and a peak of almost 1200 CCU, but 7

years after its release and the success of its sequel, the Monthly Active Users (MAU) dropped to 87. In Figure 1 the lifetime player count of *Overcooked! 2* is shown. In the last six months, it managed to get an average of 2297 MAU.

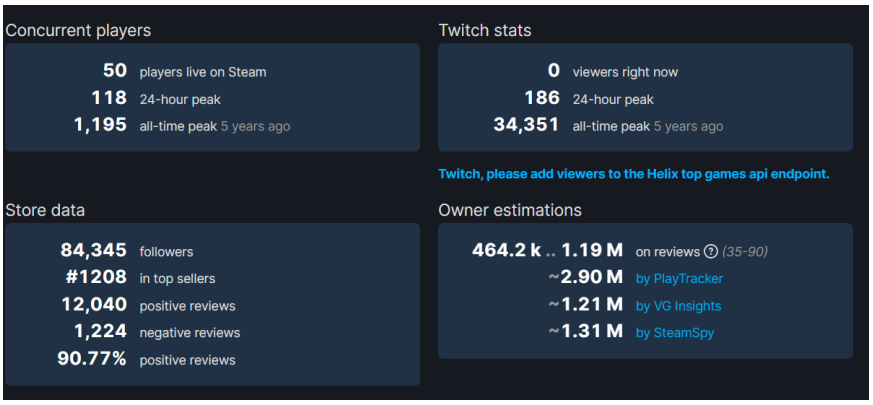


Figure 3: *Overcooked* data for SteamDB

4.2 Moving Out

Figure 4: *Moving Out* player count

Developed by SMG Studio and Devm Games and published by Team17, *Moving Out* is a co-op physics-based moving simulator where players take on moving jobs all across the busy town of Packmore. It shares a similar gameplay with *Overcooked*, although with a much lower CCU and MAU. *Moving Out* had its peak at the release date, with 652 CCU, but after a month had a huge drop to 40 MAU. Only thanks to the release of the bundle, it managed to keep an average of 100 MAU (Monthly Active Users). Currently, the game has 30 MAU (Monthly Active Users). The release of a

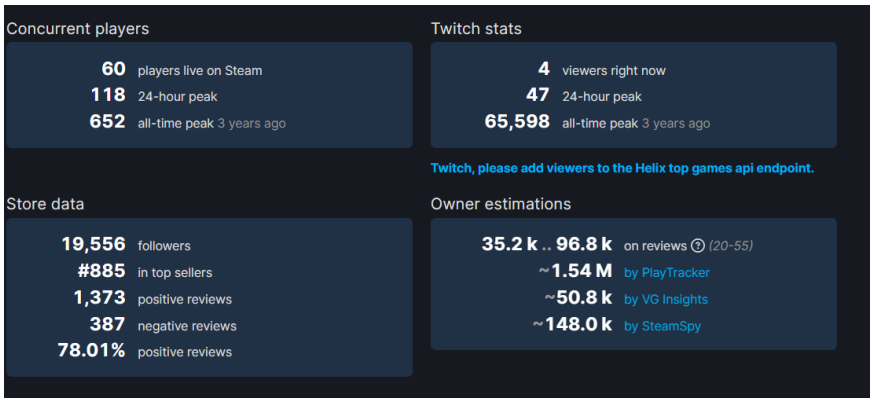
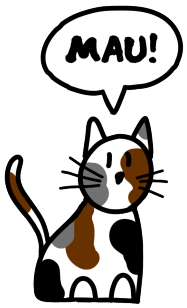


Figure 5: *Moving Out* data for SteamDB

sequel, *Moving Out 2*, has been announced. It, like *Overcooked! 2*, will be an online co-op game.

4.3 Our Estimation

Our team predicts that we will not have the same player count as the top competitor *Overcooked! 2*, mainly because it had a foundation in its prequel. Due to this, the player count estimation is based on *Overcooked*, rather than its successor, and other games. At release, the game should have between 300 to 500 CCU, and a probable peak of 1000 CCU. We also expect to be able to get an initial MAU of 500 and to stabilize at around 300 MAU. Another example to keep in mind is *Moving Out*, because it had a steep drop after its first month. Our team decided that before the final release, we shall prepare to develop at least one DLC and a launch event to keep our players engaged in the game.



5 General Architecture

The architecture for *Catnipped!* is divided into two distinct areas:

- *Front-end.* What the player will be able to interact with
- *Back-end.* The infrastructure handling the workload

In the next paragraphs, we will discuss them more in-depth.

5.1 Front-end

For the front-end, we shall provide the player access to our product through Steam for the purchase, download, and update of the game. We shall also manage a Discord server to strengthen the sense of community that we want to achieve.

Steam submission fee	100 dollars
Steam fee	30% of total revenue

Table 3: Costs for Steam

5.1.1 Steam

Steam is the external service used for distribution of our game and access to the services provided, in exchange for a submission cost and a percentage of the total revenue of the game. Steam provides a set of useful features that help the development and improvement of the product to be more organic and correct, here the major ones are summarized:

- *Steam Hubs.* Hubs enable players to share images, videos, guides, and reviews. This feature also makes it possible for developers to connect with players to answer questions and better understand what players' preferences are.
- *Updates.* Steam automatically manages updates and provides players with an easy and hassle-free method to keep the game up-to-date.
- *News.* Developers can publish news about the game that will be shown on the home page of the Steam client. This enables them to share important news with players about the game's current situation, future DLCs, and events.

In Table 3 some costs associated to the use of Steam are given.

5.1.2 Discord

Discord is a social platform that provides different ways to communicate. Through chats, voice calls, and forum-styled channels, players will be able to communicate with each other, organize lobbies and discuss game strategies. Players will also be able to communicate to developers possible problems and/or bugs. Hence, the server will be employed both as a tool to create a community and as a form of advertising. Costs for the server are highlighted in Table 4.

Discord	Free
Nitro Basic	2.99 USD/month
Nitro	9.99 USD/month

Table 4: Costs for Discord

5.1.3 Game Client

The Game Client runs in the Client machine and it's the most important front-end component. The player interacts directly with the game and, by completing a level, collects points or unlockables automatically. This data is saved in his local machine as a JSON file.

5.2 Back-end

In the back-end structure, we take advantage of several components managed by Unity Gaming Services. Through these services, we hope to strike a balance between costs and actual needs for our infrastructure, which, most likely, will not need to be able to support more than 1000 CCU. While these services will surely cover the needed functionality during development time, and for the expected amount of players at launch, if the multiplayer usage eventually exceeds the limitations of Unity Gaming Services, the connection functionality will need to be migrated to other services, such as AWS, or to a self-managed dedicated server. This is discussed more in-depth in Section 5.3

5.2.1 Relay Service

Unity Relay Service is a networking solution, hosted by Unity Technologies, that mimics the functionality of a proxy server to provide a plug & play peer-to-peer connection between players, without having to handle network ports/sockets. Relay supports DTLS (Datagram Transport Layer Security) encryption of all UDP communication passing through the Relay Server, which enables the users to change data

that is securely encrypted and authenticated. The main components of the Relay service are:

- *Relay Server.* The server acts as a proxy for the communication between the players in a game session, using low-latency datagram exchange so that players are not directly connected to each other. Relay servers are ideal for a listen-server architecture in which the hosting player acts as a server, with the other players acting as clients that join the game session. The Relay server serves as a public endpoint reachable by all players in a specific game session. This structure can address a variety of issues, namely: changing networks, IP addresses, network translation, and firewalls between players. Because the Unity Relay server acts as an intermediary between the host and the clients, IP addresses are hidden behind the server's IP address, providing a secure connection and data exchange without using any IP belonging to a player. This structure is outlined in Figure 6.

- *Allocation Service.* Before being able to interact with the Relay Server, the hosting player has to ask the Allocation Service for computational space on Unity servers to manage player data. This request is mandatory to allocate the correct amount of resources on the Relay Server for the necessary players.

- *Connection Flow.* This is the process that connects players to a Relay server. When a host player wants to create a game session, it requests an allocation from the Allocation services, which, in turn, selects an appropriate Relay Server. This request can include a specific region, or, if it is not specified, it uses QoS (Quality of Service) to select the best region for the request. When found, the Allocation service reserves space on the Relay server for the game session. At this point, the Relay server sends its IP, ports, and connection data to the host through the Allocation service. With this information, the hosting player binds to the Relay server. Then the host can request a join code from the Allocation service that allows other players to join the game session. This code can be shared through external services. When another player uses this join code, it sends a request to the Allocation service to look for the host player's allocation and returns the Relay server IP address, ports, and encrypted connection data. With this information, the joining players can bind to the host's Relay server and send a connection request to it. If successful the players can now send data through the Relay Server.

The pricing of this service is shown in Table 5.

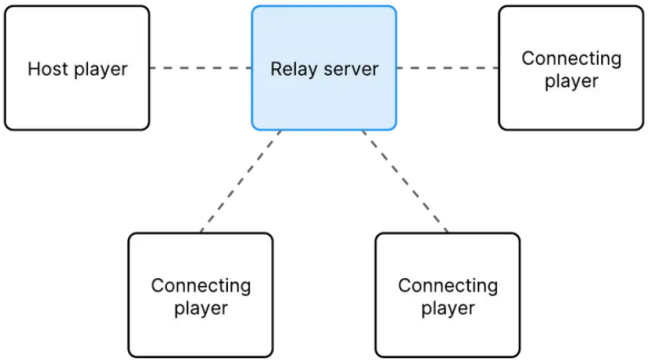


Figure 6: Connection through Relay Server

Service	Free tier	Pay as you go
CCU	50 CCU for free/month	0.16 USD per additional CCU
Bandwidth	3 GiB per CCU for free up to 150 GiB/month.	0.09 USD per GiB US + EU 0.16 USD per GiB Asia + Australia

Table 5: Costs for Relay Services

5.2.2 Lobby

Lobbies are managed by the Lobby Service, which provides an easy way to keep track of existing lobbies and enables players to browse them. Because this functionality lets players connect arbitrarily to other people's self-hosted lobbies, we adopted some security measures explained in Section 5.2.1. The Lobby Service works with the Relay service to provide a peer-to-peer connection with integrated lobbies. The game shall offer the capability to be played by users that don't have the possibility to play with friends. For this reason offering a lobby functionality is crucial. The pricing of this service can be seen in Table 6.

Service	Free tier	Pay as you go
Bandwidth	10 GiB/month US + EU and Asia + Australia	0.09 USD per GiB US + EU 0.16 USD per GiB Asia + Australia

Table 6: Costs for Relay Services

5.2.3 Authentication

The Authentication service provides an anonymous and platform-specific authentication solution for players. In our case, it is a bridge between the front-end and back-end general infrastructures (see Section 7.1). With this service, players won't have to register a game-specific account and will be able to use their Steam account (required to buy the game and use Steam client) as an online identifier. This service is free to use.

5.2.4 Analytics

Analytics is a service that provides end-to-end data analysis designed to support the game and its team. This service lets us understand and get data on players' behaviors and the game's performance. For example, it can give us information about how many people start playing at launch, every day, and in a given month. The core metrics monitored:

- New installs
- Daily active users (DAU)
- Monthly active users (MAU)
- Days Played in the last 7 Days
- Total Days Played
- Total Sessions Played
- Average Minutes Played Per Day in the Last 7 Days
- Players Categorization

With all of this information, we can also utilize the Remote Config feature to change some variables in the game in order to obtain different effects, for example:

- Adjusting Gameplay
- Enabling or Disabling Features
- Changing Graphical Themes.

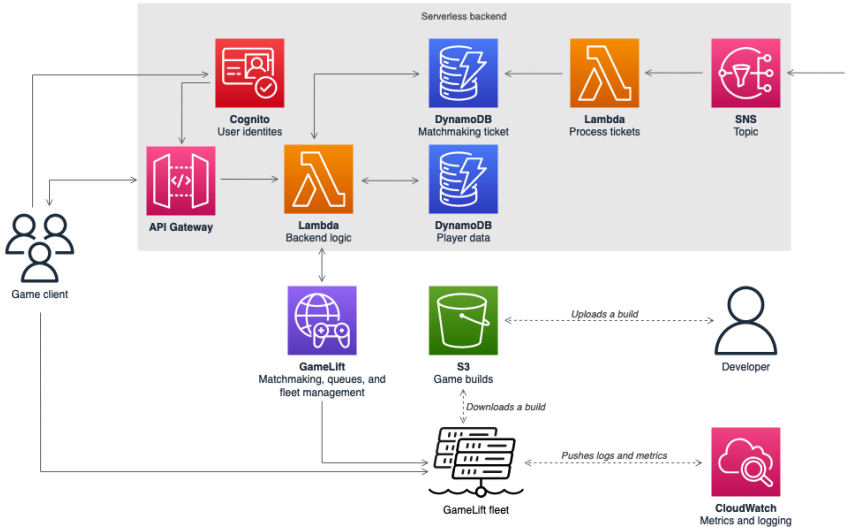


Figure 7: GameLift with serverless back-end

After making these changes with Remote Config, every client starting a new game session downloads the new configuration values, with no need to update the game. This service collects personally identifiable information and follows the EU's General Data Protection Regulation (GDPR) compliance, for more information please refer to `https://docs.unity3d.com/Manual/UnityAnalyticsDataPrivacy.html`. For this reason, we're required to integrate into the game a feature that allows the player to opt out of the information collection and personal data management from Unity Analytics. The pricing of this service can be seen in Table 7.

Service	Free tier	Pay as you go
Events	25,000,000 events/month	0.0027 USD per 1000 events
SQL Data Explorer	2000 query/second per month	0.0058 USD per second
Remote Settings	free of charge	free of charge

Table 7: Costs related to the Analytics Service

5.3 AWS

The current implementation of the game can manage the connection workload of development and estimated MAU after launch. However, at any time, if the system fails to provide a reliable and functional service, we evaluated a possible solution for

the product, able to scale to a higher player count and manage more traffic: AWS, specifically GameLift. It is important to keep in mind that this represents a fallback solution, not the one that is actually being implemented. AWS offers a collection of services that, depending on the type of system we need to implement, will ensure that network performance is on par with the rest of the game. The following components will be considered, if needed, for the task.

- *GameLift.* Service used to deploy dedicated servers in the cloud for session-based multiplayer. Its main objective is to manage fleets, game sessions, and matchmaking. A fleet in Gamelift represents the hosting resources as a set of Amazon Elastic Compute Cloud (EC2) instances, which are web services (virtual machines) that represent a physical server where the application is deployed.

- *Amazon Cognito.* Service that provides authentication, authorization, and user management. The user is able to automatically sign in through a third party, in our case Steam.

- *Amazon SNS.* Amazon Simple Notification Service is a service managed by Amazon that provides notification delivery for events. Gamelift, for example, sends an event notification to clients through SNS whenever it places a session in one of the fleet locations.

- *Amazon API Gateway.* Service used for creating, publishing, maintaining, monitoring, and securing REST, HTTP, and WebSocket APIs.

- *Amazon DynamoDB.* Service that can create database tables able to store and retrieve any amount of data and serve any level of request traffic. It's used to store player data and matchmaking information.

- *AWS Lambda.* The service that lets us run code without provisioning or managing a server. It's the central point of the back-end infrastructure, communicating with Amazon GameLift for matchmaking and Amazon DynamoDB for player data.

- *Amazon S3.* Amazon's simple storage service is a service used for cloud Storage, in particular for the images of the game server that can be downloaded by the GameLift fleet.

- *Cloud Watch.* A service that monitors your application in real time in order to tune and resolve problems in the server fleets.

Pricing with AWS free tier can be seen in Table 8. We estimated the price of using GameLift, considering a game like *Overcooked! 2*, using the tool "AWS Pricing Calculator" (accessible here). The result would be a monthly cost of 2.022,31 USD.

Service	Content	Price
Amazon Cognito	50.000 MAU 10GB cloud archive 1 million synchronization operation	12 months free
Amazon SNS	1 million publications 100.000 HTTP/s send 1.000 e-mail send	free
Amazon DynamoDB	25Gb of storage 200 million request	free
API Gateway	1 million incoming API calls/month	12 months free
Lambda	1 million free requests per month 3.2 million seconds of processing time	free
Amazon S3	5 GB of storage 20.000 GET requests 2.000 PUT requests	12 months free

Table 8: Costs associated to AWS free tier

5.4 Workload Capacity

Following the estimation presented in Section 4.3, the front-end and back-end infrastructures are estimated to be able to handle the estimated workload.

5.4.1 Front-end

On the front-end side, we do not expect problems regarding Steam, which is a third-party service, or the Discord server.

5.4.2 Back-end

Unity Services can handle the estimated workload that we analyzed in Section 4.3, thanks to the backbone of our infrastructure, Unity Relay Service, which will be used to manage scaling and communication between players in a game session. Unity is responsible for maximizing CPU and network usage to meet user demand. This solution solves the problem of firewalls, open ports, and sharing IP addresses, with the added benefit of avoiding the higher costs of implementing a dedicated server.

6 External Services

In this section, the external services we will use are presented, and costs associated with some of them are given in Section 10.

6.1 Advertisement on social media

Catnipped! is a game that relies both on being a funny game to play with friends and on the comedy and irony of its gameplay elements. For these reasons, the advertisement would be more effective if done both on social media, with particular attention to TikTok, Twitch, and Youtube, and by gifting game keys to streamers. Games like *Overcooked! 2* or *Moving Out* have generated lots of interest thanks to streams, in Figure 8 the number of Twitch viewers for *Overcooked! 2* can be seen.

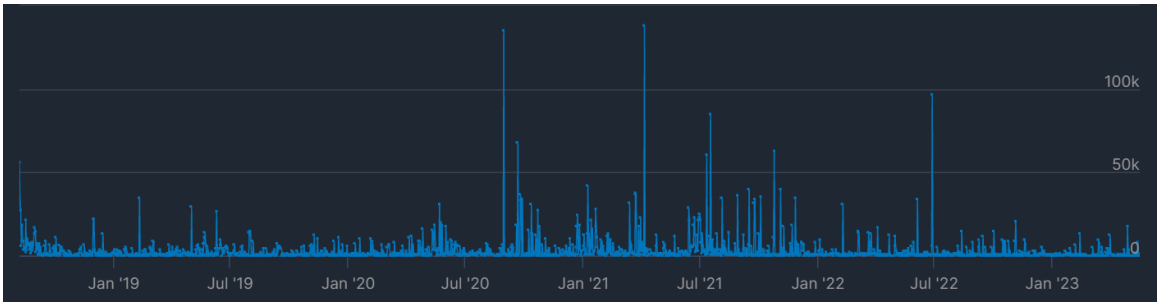


Figure 8: Twitch Viewers for *Overcooked! 2*

6.2 Payments

After the game is published, Steam will manage the whole payment system, including sales, discounts, and refunds. Steam will also automatically apply fees on game revenues.

6.3 Achievements & Leaderboards

Catnipped! relies heavily (but not solely) on achievements to unlock new skins, new levels, and collectibles, while keeping players interested. For this reason, the game will implement a set of Steam achievements, visible through the client, and some collectibles inside the game, not listed in the achievements. Every level will also have a leaderboard to keep track of the best score for that specific level.

6.4 Authentication

Catnipped! shall implement a multiplayer system based on lobbies. consequently, players will automatically authenticate through their Steam account (necessary to buy the game and use the client) and use their Steam profile name as the in-game name.

6.5 Legal support for terms of service

Our team will need a legal support professional figure to help with the writing of the GDPR agreement for analytics data usage, terms of service, and contractor agreements. We will also need help trademarking the name of the game and managing the theme of drug usage inside the game.

7 Connection

This section shows how our general game infrastructure interfaces with the player. The back-end is fully managed by Unity Gaming Services, while the front-end manages the ones offered to the player. For work-office connection, we rely on an internet plan provided by the internet provider.

7.1 Global Infrastructure

Figure 9 shows how all the services discussed in Section 5 communicate with each other. While the Relay service can provide different server regions, we will focus on providing the client with the three major ones (Europe, America, and Asia) in order to reach as many players as possible and reduce latency during gameplay.

7.2 Network Requirements

Catnipped!, being a light game, does not need a big amount of data to work. The most crucial elements needed to guarantee an optimal game experience are:

- *Network.* The client's minimum required connection shall be at least 10 Mbps, with a recommended minimum connection of 20 Mbps.
- *Bandwidth usage.* By analyzing the market, specifically *Overcooked! 2*, we predict that our game is going to consume at least an average of 50 MB of data per hour for a continuous gaming session.

- *Latency.* With a latency of around 100 ms, the game shall still be playable with minor lag, any more than that could ruin the immersion of the game for players.
- *Type of Connection.* For our office, we will require at least an ADSL/SDLS broadband service and we think that the speed required for our workplace should be at least 20 Mbps.

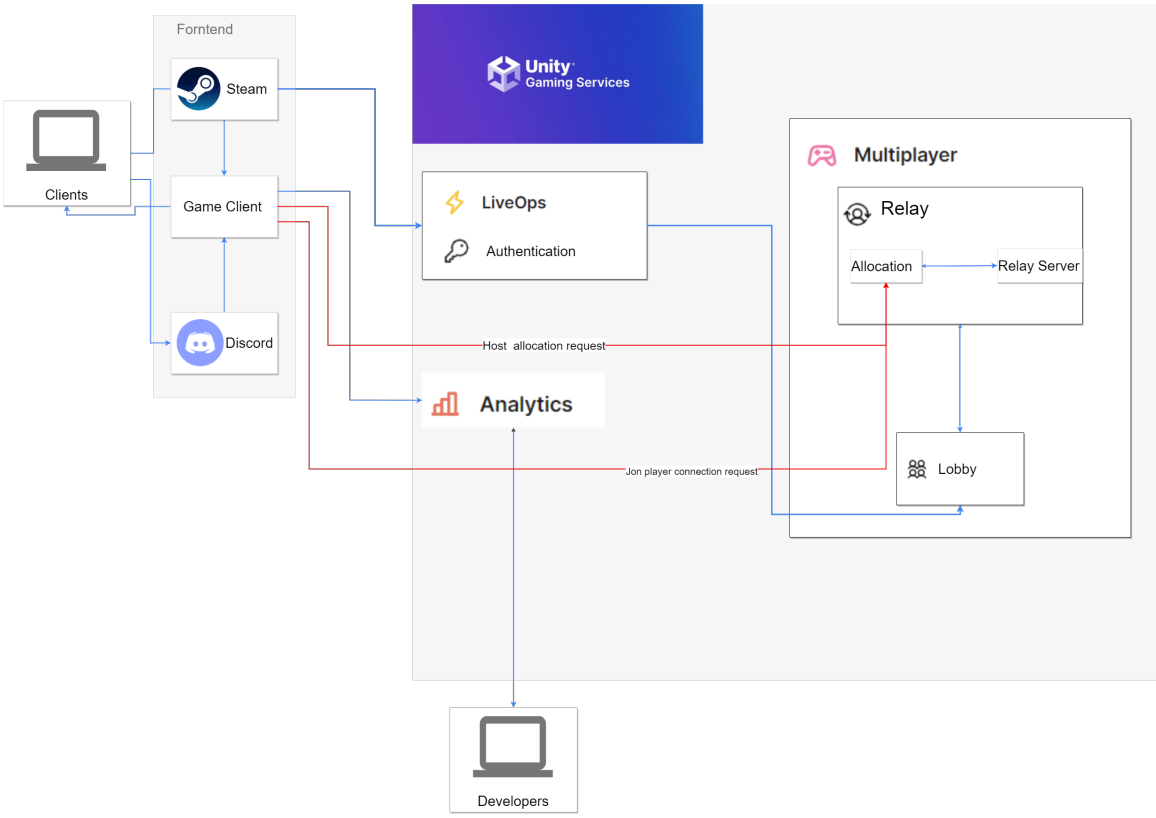
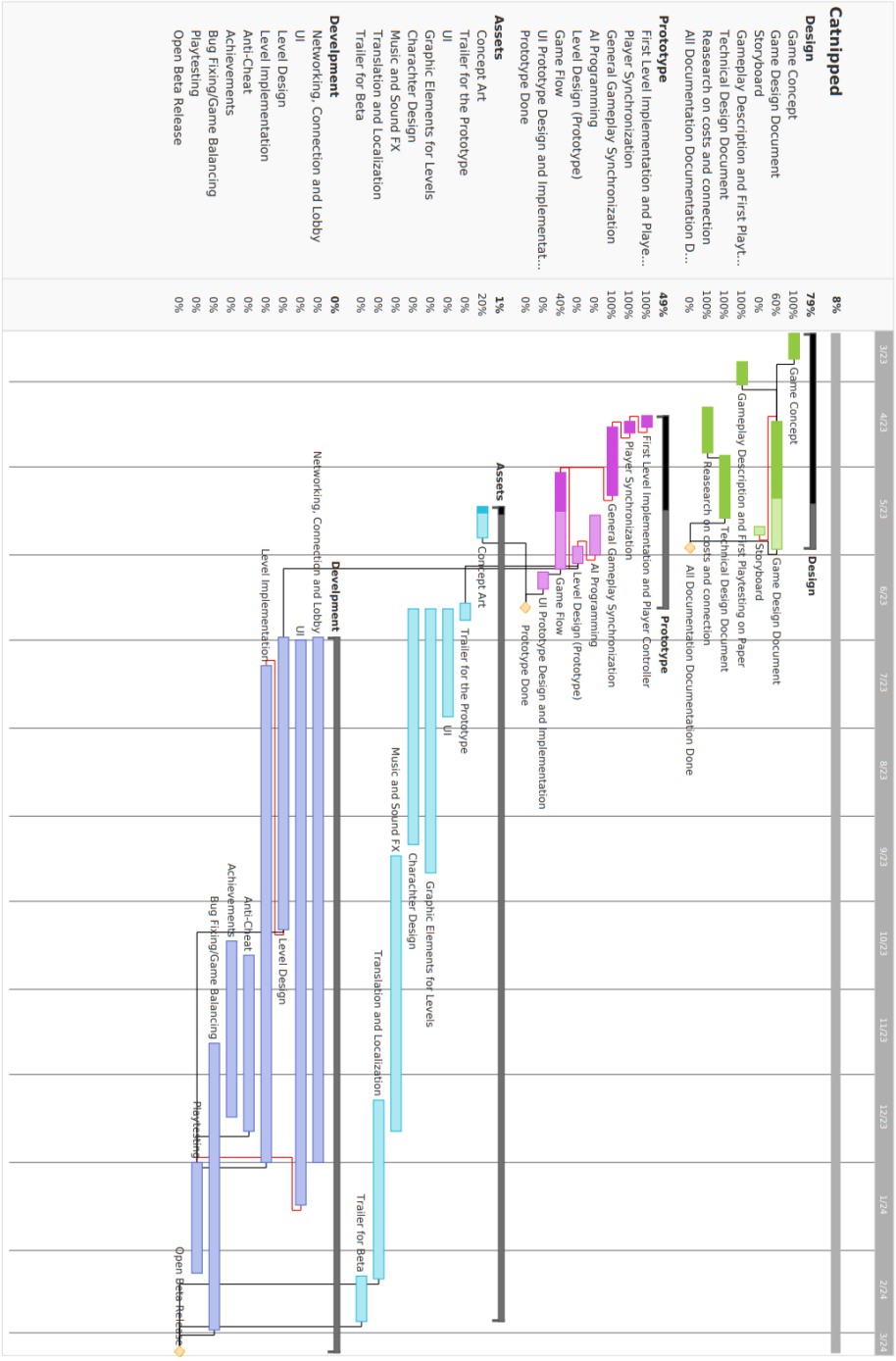


Figure 9: Game Infrastructure

8 Gantt


Catnipped
8%
Design
79%
Game Concept
100%
Game Design Document
60%
Storyboard
0%
Gameplay Description and First Playt...
100%
Technical Design Document
100%
Reasearch on costs and connection
100%
All Documentation Documentation D...
0%
Prototype
49%
First Level Implementation and Playe...
100%
Player Synchronization
100%
General Gameplay Synchronization
100%
AI Programming
0%
Level Design (Prototype)
0%
Game Flow
40%
UI Prototype Design and Implementat...
0%
Prototype Done
0%
Assets
1%
Concept Art
20%
Trailer for the Prototype
0%
UI
0%
Graphic Elements for Levels
0%
Charachter Design
0%
Music and Sound FX
0%
Translation and Localization
0%
Trailer for Beta
0%
Develpment
0%
Networking, Connection and Lobby
0%
UI
0%
Level Design
0%
Level Implementation
0%
Anti-Cheat
0%
Achievements
0%
Bug Fixing/Game Balancing
0%
Playtesting
0%
Open Beta Release
0%
3/23
4/23
5/23
6/23
7/23
8/23
9/23
10/23
11/23
12/23
1/24
2/24
3/24
Design
Game Concept
Game Design Document
Storyboard
Gameplay Description and First Playtesting on Paper
Technical Design Document
Reasearch on costs and connection
All Documentation Documentation Done
Prototype
First Level Implementation and Player Controller
Player Synchronization
General Gameplay Synchronization
AI Programming
Level Design (Prototype)
Game Flow
UI Prototype Design and Implementation
Prototype Done
Assets
Concept Art
Trailer for the Prototype
UI
Graphic Elements for Levels
Charachter Design
Music and Sound FX
Translation and Localization
Trailer for Beta
Develpment
Networking, Connection and Lobby
UI
Level Design
Level Implementation
Anti-Cheat
Achievements
Bug Fixing/Game Balancing
Playtesting
Open Beta Release

9 Delivery

9.1 Estimated Delivery Time

Our team expects to deliver the first prototype in the second half of June 2023, with a few levels working and a demonstration of the main game mechanics, like building and farming. Our road map includes:

- First open Beta release in 10 months
- Final release of the game in 11 months
- First DLC after 4 months from the final release

For a more in-depth roadmap, please refer to Section 8.

9.2 Delivery Platform

The chosen delivery platform for our game is Steam: this makes it possible to have the open Beta and the final product using the same product page in the Steam Shop. Also, Steam makes it easy for developers to publish future DLCs and add paid content like event-themed skins. This includes Halloween, Christmas, Easter, and similar events that usually create the opportunity for the community to play with friends or relatives.

9.3 Delivery Methodology

Our game shall have an open beta release and a full release. The open beta release shall be free to play: this will help make the game more appealing to all kinds of players and streamers that will spread the word about the game, while also giving useful information to developers. The final release will be paid for and priced at around 10 USD. As a launch offer, the game could be discounted by 20%, down to 7.99$, for players who played at the open beta.

9.4 Post launch

The development team expects to have future DLCs, events, and updates including new levels, skins, and possibly mechanics. This is done to keep players interested in the game for a longer period of time to sustain development costs. Additional content can be categorized into two classes:

- *Generic.* Some DLCs can be themed around generic themes, like environments (jungle, mountain, pirates, or Vikings).
- *Holidays.* We can also have in-game events based on holidays throughout the year, for example with Christmas or Easter themed skins and levels.

It should also be noted that constant bug-fixing updates are required to keep the game polished and up-to-date.

10 Cost Estimation

In this section, an estimate of the costs we will need to sustain is given. In Section 10.7 the total costs are summed up.

10.1 Team

Profession	Amount	Salary
Game Designer	2	2500 USD/month
Game Programmer	1	2500 USD/month
Social Media Manager	1	1300 USD/month
Graphic Designer	1	2000 USD/project
3D Artist & Animator	1	2000 USD/project
Composer	1	1500 USD/project
Legal Assistant	1 x 200 hours	30 USD/hour
Total Monthly Cost		8800 USD/month
Total Yearly Cost		105600 USD/year

10.2 Location

Service	Amount	Price
Office rent	1	1500 USD/month
Desk	6	140 USD
Chairs	6	160 USD
Lights	10	20 USD
Drawing board	1	35 USD
Bulletin board	1	35 USD
Internet	1	30 USD/month
Electricity	-	150 USD/month
Cleaning service	Every week	150 USD/month
Total one time		2070 USD
Total monthly		1830 USD/month
Total yearly		21960 USD/year

10.3 Hardware

Hardware	Amount	Price
Laptop	6	1500 USD
Graphic Tablet	1	500 USD
Total		9500 USD

10.4 Software

Software	Licenses	Price
Microsoft 365 Business Basic	5	6 USD/month
Adobe Suite	2	80 USD/month
Unity Pro	3	190 USD/month
Visual Studio Professional	1	45 USD/month
Blender	2	0 USD/month
Total monthly		805 USD/month
Total yearly		9660 USD/year

10.5 Unity Gaming Service

The Unity services, mentioned in Section 5.2, have a free tier and we expect not to exceed these limits during the development, since only the development team and later some beta testers will play the game.

10.6 Advertisement

As explained in Section 6.1, we believe that *Catnipped!* would get a greater audience with advertisements on social media and streaming platforms. For this reason, a Social Media Manager will be needed, the salary expected for the SMM is given in Section 10.1. While the cost of gifting keys to streamers can't be measured directly, the following table sums up the cost of advertisement on various social media platforms in CPM (Cost Per Mille impressions). These prices are averages because the exact price depends on the performance of the ad campaign.

Platform	**Price (CPM)**
TikTok	10 USD
Facebook	12 USD
Twitter	7 USD
Instagram	7 USD
Twitch	3.50 USD
Youtube	2 USD

10.7 Total

To sum up the costs for the production and development of *Catnipped!* we arranged a simple summary in Table 9, completed with the budget we ask for, which will cover all costs until the release of the first version of the game. Residues will be used to cover one time costs, single projects, and other expenses, such as marketing costs.

	Price
Team	105600 USD/year
Location	21960 USD/year
Hardware	9500 USD
Software	9660 USD/year
Total one time	11570 USD
Total monthly	11450 USD/month
Total yearly	137220 USD/year
Budget	160000 USD/year

Table 9: Total production cost